4.5. **Wasserstein exponential decays.** Assume that the Lyapunov functions (ψ,φ) defined in (4.23) satisfy the estimates

$$e^{\delta U(x_1)}+e^{\delta U(x_2)}\geqslant \varrho_\varphi(x_1,x_2)\quad\text{and}\quad e^{\delta V(y_1)}+e^{\delta V(y_2)}\geqslant \varrho_\psi(y_1,y_2) \tag{4.27}$$

for some semi-metric $\varrho_\varphi,\varrho_\psi$. In this case, by (1.10) there exists $c_u,c_v>0$ such that

$$\mathbb{W}_{\varrho_\varphi}(\pi_{2n+1},\lambda_U)\leqslant c_u\ \|\!|\pi_{2n+1}-\lambda_U|\!\|_\varphi\quad\text{and}\quad \mathbb{W}_{\varrho_\psi}(\pi_{2n},\nu_V)\leqslant \|\!|\pi_{2n}-\nu_V|\!\|_\psi$$

with the Wasserstein semi-metrics $\mathbb{W}_{\varrho_\varphi},\mathbb{W}_{\varrho_\psi}$ defined in (1.8). This yields the following Wasserstein estimates.

Corollary 4.11. *Assume $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta\in]0,1]$. In addition, the Lyapunov functions $(e^{\delta U},e^{\delta V})$ satisfy (4.27) for some semi-metric $(\varrho_\varphi,\varrho_\psi)$. In this situation, there exists some $n_0\geqslant 0$ and $c_u,c_v\geqslant 0$ such that for any $n\geqslant n_0$ we have*

$$\begin{aligned}\mathbb{W}_{\varrho_\psi}(\pi_{2n},\nu_V)&\leqslant c_v\ \rho^{n-n_0}\ \|\!|\pi_{2n_0}-\nu_V|\!\|_\psi\\ \mathbb{W}_{\varrho_\varphi}(\pi_{2n+1},\lambda_U)&\leqslant c_u\ \rho^{n-n_0}\ \|\!|\pi_{2n_0+1}-\lambda_U|\!\|_\varphi,\end{aligned} \tag{4.28}$$

In the above display, $n_0\geqslant 1$ and $\rho\in[0,1[$ are as in (4.25) and (ψ,φ) are the Lyapunov functions that depend on δ defined in (4.23).

Assume there exists some semi-metric ϱ_U,ϱ_V and some constants $c_u,c_v\in\mathbb{R}$ such that

$$\begin{aligned}U(x_1)+U(x_2)&\geqslant c_u+\varrho_U(x_1,x_2),\\ V(y_1)+V(y_2)&\geqslant c_v+\varrho_V(y_1,y_2).\end{aligned} \tag{4.29}$$

In this case, by (3.5), for any $p\geqslant 1$ there exists some constant $c_p>0$ such that

$$\mathbb{W}_{p,\varrho_U}(\pi_{2n+1},\lambda_U)^p\leqslant c_p\ \|\!|\pi_{2n+1}-\lambda_U|\!\|_\varphi\quad\text{and}\quad \mathbb{W}_{p,\varrho_V}(\pi_{2n},\nu_V)^p\leqslant c_p\ \|\!|\pi_{2n}-\nu_V|\!\|_\psi,$$

with the p-Wasserstein semi-metrics $\mathbb{W}_{p,\varrho_U},\mathbb{W}_{p,\varrho_V}$ defined in (1.8). This yields the following Wasserstein estimates.

Corollary 4.12. *Assume $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta\in]0,1]$. In addition, the target potential functions (U,V) satisfy (4.29) for some semi-metric (ϱ_U,ϱ_V). In this situation, for any $p\geqslant 1$ there exists some $c_p\geqslant 0$ such that for any $n\geqslant n_0$ we have*

$$\begin{aligned}\mathbb{W}_{p,\varrho_V}(\pi_{2n},\nu_V)^p&\leqslant c_p\ \rho^{n-n_0}\ \|\!|\pi_{2n_0}-\nu_V|\!\|_\psi,\\ \mathbb{W}_{p,\varrho_U}(\pi_{2n+1},\lambda_U)^p&\leqslant c_p\ \rho^{n-n_0}\ \|\!|\pi_{2n_0+1}-\lambda_U|\!\|_\varphi,\end{aligned} \tag{4.30}$$

In the above display, $n_0\geqslant 1$ and $\rho\in[0,1[$ are as in (4.25) and (ψ,φ) are the Lyapunov functions that depend on δ defined in (4.23).

Remark 4.13. *Using Proposition D.3 in Appendix D and following exactly the same argument as in the proof of Theorem 4.8, the estimate (4.24) remains valid when the weaker condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ holds for some $n_0\geqslant 0$, provided that*

$$\left(\|\frac{d\pi_{2n_0}}{d\nu_V}\|_\psi\vee\|\frac{d\nu_V}{d\pi_{2n_0}}\|_\psi\right)\vee\left(\|\frac{d\pi_{2n_0+1}}{d\lambda_U}\|_\varphi\vee\|\frac{d\lambda_U}{d\pi_{2n_0+1}}\|_\varphi\right)<\infty.$$

4.6. **Entropy exponential decays.** Let us discuss some fairly straightforward consequences of the estimate (4.24). First note that (4.24) ensures the convergence of the series

$$\begin{aligned}\lim_{n\to\infty}U_{2n}=\mathbb{U}&:=U_0+\sum_{p\geqslant 0}\log\frac{d\pi_{2p+1}}{d\lambda_U}\quad\text{in }\mathcal{B}_\varphi(\mathbb{X}),\\ \lim_{n\to\infty}V_{2n}=\mathbb{V}&:=V_0+\sum_{p\geqslant 0}\log\frac{d\pi_{2p}}{d\nu_V}\quad\text{in }\mathcal{B}_\psi(\mathbb{Y}),\end{aligned} \tag{4.31}$$

where $U_0 = U$ and $V_0 = 0$. Applying Lebesgue's dominated convergence theorem, the integral equations stated in (2.4) converge as $n \to \infty$ to a system of integral equations

$$\mathbb{U} = U + \log \mathcal{Q}(e^{-\mathbb{V}}) \quad \text{and} \quad \mathbb{V} = V + \log \mathcal{R}(e^{-\mathbb{U}}).$$

More precisely, using the estimates in (4.24) it is straightforward to prove the following corollary.

Corollary 4.14. *Under the assumptions of Theorem 4.8 for any $n \geqslant 1$ we have*

$$\|\mathbb{U} - U_{2n}\|_\psi \vee \|\mathbb{V} - V_{2n}\|_\varphi \leqslant \frac{c}{1-\rho^2}\ \rho^{2n} \tag{4.32}$$

with the constant $c > 0$ as in (4.24). In addition, we have the relative entropy estimates

$$\mathrm{Ent}\,(\lambda_U \mid \pi_{2n+1}) \leqslant c\ \lambda_U(\varphi)\ \rho^{2n} \quad \text{and} \quad \mathrm{Ent}\,(\nu_V \mid \pi_{2n}) \leqslant c\ \nu_V(\psi)\ \rho^{2n}. \tag{4.33}$$

With this notation at hand, the bridge distribution (1.4) has the form

$$\lim_{n\to\infty} \mathcal{P}_n(d(x,y)) = \mathbb{P}(d(x,y)) := e^{-W(x,y)}\ \lambda_{\mathbb{U}}(dx)\ \nu_{\mathbb{V}}(dy). \tag{4.34}$$

Moreover, we have

$$\begin{aligned}\mathrm{Ent}(\mathbb{P} \mid \mathcal{P}_{2n}) &= \lambda_U(U_{2n} - \mathbb{U}) + \nu_V(V_{2n} - \mathbb{V}) \\ &= \lambda_U\left(\sum_{q\geqslant n} \log \frac{d\lambda_U}{d\pi_{2q+1}}\right) + \nu_V\left(\sum_{q\geqslant n} \log \frac{d\nu_V}{d\pi_{2q}}\right).\end{aligned}$$

The exponential estimate (4.24) also justifies the interchange of summation and integration. The next corollary is now a direct consequence of the relative entropy estimates in (4.33).

Corollary 4.15. *Under the assumptions of Theorem 4.8 we obtain*

$$\mathrm{Ent}(\mathbb{P} \mid \mathcal{P}_{2n}) = \sum_{p\geqslant n} \mathrm{Ent}\,(\lambda_U \mid \pi_{2p+1}) + \sum_{p\geqslant n} \mathrm{Ent}\,(\nu_V \mid \pi_{2p}) \quad \textit{for any } n \geqslant 1.$$

In addition, there exists some $c > 0$ such that, for any $n \geqslant 1$, we have the entropy exponential decay estimates

$$\mathrm{Ent}(\mathbb{P} \mid \mathcal{P}_{2n}) \leqslant c\ \rho^{2n}\ (\lambda_U(\varphi) + \nu_V(\psi)).$$

5. Some illustrations

In this section, we illustrate the scope and flexibility of the theoretical results developed above through a collection of representative examples. We show how the Lyapunov, drift-minorization, and weighted contraction conditions can be verified in a variety of settings, ranging from polynomial and heavy-tailed models to boundary-driven, semi-compact, and Gaussian frameworks. These examples highlight how the abstract assumptions translate into explicit growth, integrability, and regularity conditions on the target potentials and transition kernels.

5.1. **Polynomial growth models.** Consider some finite dimensional normed vector space $\mathbb{X} = \mathbb{Y}$ equipped with measure $\lambda = \nu$ such that for any $p > 0$ and $\tau > 0$ we have

$$\int e^{-\tau\|x\|^p}\ \lambda(dx) < \infty. \tag{5.1}$$

In this context, we have $\lambda_U(dx) = e^{-U(x)}\ \lambda(dx)$ and $\nu_V(dx) = \lambda_V(dx) = e^{-V(x)}\ \lambda(dx)$.

Assume that the functions U, V and W_U, W^V are continuous. In addition, there exists some parameters $c_w, c_u, c_v \in \mathbb{R}$ as well as $\varsigma_u, \varsigma_v, \tau_u, \tau_v, p_u, p_v > 0$ and $q_u, q_v \geqslant 0$ such that

$$\begin{aligned} W(x,y) &\leqslant c_w + \varsigma_u\ \|x\|^{p_u} + \varsigma_v\ \|y\|^{p_v}, \\ U(x) &\geqslant c_u + \tau_u\ \|x\|^{p_u+q_u} \quad \text{and} \quad V(x) \geqslant c_v + \tau_v\ \|x\|^{p_v+q_v}. \end{aligned} \tag{5.2}$$

Proposition 5.1. *Condition $\mathcal{H}'_\delta(U,V)$ is satisfied for some $0<\delta<1/2$ whenever, for any parameter $\iota\in\{u,v\}$, one of the following assertions is satisfied*

$$(q_\iota>0)\quad \text{or}\quad (q_\iota=0\quad \text{and}\quad \tau_\iota/2>\varsigma_\iota)\,. \tag{5.3}$$

Proof. The exponential moment condition (5.1) ensures that (4.21) holds for any $0<\delta<1/2$ and the probability measures λ_U and λ_V have finite p_u and p_v absolute moments

$$m_u^{p_u}:=\int\|x\|^{p_u}\lambda_U(dx)<\infty\quad\text{and}\quad m_v^{p_v}:=\int\|x\|^{p_u}\lambda_V(dx)<\infty.$$

Also note that

$$\mathcal{U}_\delta(x)=\delta U(x)-W^V(x)\quad\geqslant\quad(\delta c_u-c_w-\varsigma_v\,m^{p_v})+\delta\tau_u\,\|x\|^{p_u+q_u}-\varsigma_u\,\|x\|^{p_u}.$$

This shows that $\mathcal{U}_\delta$ has compact sub-level sets with $0<\delta<1/2$ as soon as

$$q_u>0\quad\text{or}\quad(q_u=0\quad\text{and}\quad\delta\tau_u>\varsigma_u)\,.$$

In the same vein, condition (5.3) with $a=v$ ensures $\mathcal{V}_\delta$ has compact sub-level sets. ∎

For instance, for bi-Laplace transitions on $\mathbb{X}=\mathbb{R}=\mathbb{Y}$ we have

$$\mathcal{Q}(x,dy)=\frac{\varsigma}{2}\,e^{-\varsigma|x-y|}\,dy\quad\text{with }\varsigma>0$$

$$\Longrightarrow W(x,y)=c_w+\varsigma|x-y|\leqslant c_w+\varsigma|x|+\varsigma|y|\quad\text{with }c_w:=\log(2/\varsigma)$$

$$\Longrightarrow|W(x,z_1)-W(x,z_2)|\vee|W(z_1,y)-W(z_2,y)|\leqslant\varsigma\,|z_1-z_2|.$$

For linear Gaussian models, the r.h.s. condition in (5.3) is satisfied as soon as the covariance of the linear Gaussian transition $\mathcal{Q}$ is defined by (1.2) is sufficiently large. In the same context of linear and Gaussian models, we show in Proposition 5.3 that this condition is in fact equivalent to condition $\mathcal{H}_\delta(U,V)$.

By (5.2) there exists some constant $\overline{\tau}_u>0$ such that

$$U(x_1)+U(x_2)\quad\geqslant\quad\overline{c}_u+\overline{\tau}_u\,(\|x_1\|+\|x_2\|)^{p_u+q_u}\quad\text{with}\quad\overline{c}_u:=2c_u.$$

This yields the estimate

$$U(x_1)+U(x_2)\geqslant\overline{c}_u+\varrho_U(x_1,x_2)\quad\text{with}\quad\varrho_U(x_1,x_2):=\overline{\tau}_u\,\|x_1-x_2\|^{p_u+q_u}.$$

Similarly, there exist some constants $\overline{c}_u\in\mathbb{R}$ and $\overline{\tau}_v>0$ such that

$$V(x_1)+V(x_2)\geqslant\overline{c}_v+\varrho_V(x_1,x_2)\quad\text{with}\quad\varrho_V(x_1,x_2):=\overline{\tau}_v\,\|x_1-x_2\|^{p_v+q_v}.$$

We conclude that (U,V) satisfy (4.29) with the semi-metric (ϱ_U,ϱ_V) so that the Wasserstein estimates (4.30) hold.

5.2. **Heavy tailed transitions.** Consider the normed vector space $\mathbb{X}=\mathbb{Y}$ and the reference measure $\lambda=\nu$ discussed in Section 5.1. Assume some parameter $c_w\in\mathbb{R}$, $\tau_{p,w},\tau_{q,w}>0$ and $p_w\wedge q_w>1$ such that

$$W(x,y)\leqslant c_w+\log\left(1+\tau_{p,w}\,\|x\|^{p_w}+\tau_{q,w}\,\|y\|^{q_w}\right).$$